ReadmitGuard Analytics

Business Requirements Document (BRD)

30-Day Readmission Risk Dashboard

Project Title: 30-Day Readmission Risk Dashboard
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1. Executive Summary

This project analyzes historical patient encounter data to identify the key factors driving 30-day hospital readmissions. The analysis will combine Excel-based exploration, SQL querying, and Power BI visualization to uncover readmission trends across patient demographics, diagnoses, and treatment patterns. The resulting dashboard will give healthcare administrators and clinical teams a clear, data-driven view of high-risk patient segments to support proactive care planning and resource allocation.

2. Business Objectives

- Identify the key risk factors most strongly associated with 30-day patient readmissions.
- Visualize readmission distribution by diagnosis, age group, gender, and race.
- Quantify readmission rates across admission types, payer codes, and medical specialties.
- Enable early identification of high-risk patients to support proactive care interventions.
- Deliver an interactive Power BI dashboard that supports healthcare administrators and clinical teams in resource and policy decisions.

3. Background and Current State

Patient encounter data is currently captured for billing and clinical documentation purposes but is not systematically analyzed for readmission risk. Reporting is largely retrospective and manual, with no consolidated view linking patient demographics, diagnoses, medications, and length of stay to readmission outcomes. This limits the hospital's ability to identify high-risk patients proactively and makes it difficult to target interventions or evaluate the effectiveness of discharge planning.

A structured analytics approach is needed to consolidate this data, surface readmission patterns, and present findings in a format that supports action rather than just historical reporting.

4. Scope

In Scope:

- Analysis of historical, anonymized patient encounter data (diabetic_data.csv).
- Exploratory data analysis and pivot-based reporting in Excel.
- Schema creation, data loading, and query-based analysis in SQL.
- Development of an interactive Power BI dashboard with demographic, diagnosis, and outcome visuals.
- Use of AI-assisted features (Excel =AI.TABLE, ChatGPT for Excel, Power BI Quick Measures) to accelerate insight generation.

Out of Scope:

- Integration with live hospital EHR/PMS systems or real-time data feeds.
- Clinical decision-making, diagnosis, or treatment recommendations.
- Development or deployment of a predictive machine learning model.
- Handling of personally identifiable patient information.

5. Stakeholders

Role	Name / Dept	Responsibility
Project Sponsor	Dr. Amanda Clark	Approves scope, budget, and strategic alignment
Business Analyst	[Your Name]	Gathers requirements; performs analysis and builds the dashboard
Hospital Administrator	Operations	Uses insights for resource and staffing planning
Compliance Officer	Legal / Compliance	Ensures adherence to data privacy standards (e.g., HIPAA)
Business Intelligence Analyst	Analytics Team	Reviews dashboards and validates visual accuracy

6. Business Requirements

ID	Requirement Description	Priority	Source
BR001	Patient encounter, diagnosis, and discharge data must be merged into structured SQL tables for querying.	High	Business Analyst
BR002	The dashboard must visualize readmission distribution by age group, gender, and diagnosis.	High	Hospital Administrator
BR003	The analysis must calculate readmission rates across admission types and payer codes.	High	Compliance Officer

ID	Requirement Description	Priority	Source
BR004	AI-assisted features must generate readmission risk summaries by age group and flag high-risk segments.	Medium	Business Analyst
BR005	The dashboard must correlate medications, procedures, lab tests, and length of stay with readmission status.	Medium	BI Analyst
BR006	All analysis must use anonymized data only, with no patient-identifiable information exposed.	Critical	Legal / Compliance
BR007	Deliverables must be built using Excel, SQL, and Power BI only, without external system integration.	High	IT / Tooling Constraint

7. Assumptions

- The provided dataset (diabetic_data.csv) is representative and sufficiently complete for meaningful analysis.
- Data has already been anonymized and contains no personally identifiable information.
- The analyst has working access to Excel, a SQL environment, and Power BI Desktop.
- Key stakeholders will be available for periodic review of findings and the dashboard.

8. Constraints

- Analysis is based on historical, retrospective data only — no real-time data feed is available.
- Tooling is limited to Excel, SQL, and Power BI; no additional software budget is available.
- The project must be completed within the capstone course timeline.
- No integration with live hospital systems is permitted during this phase.

9. Risks

Risk ID	Description	Likelihood	Impact	Mitigation Plan
R-001	Data quality issues (missing values, placeholder entries such as '?').	Medium	Medium	Clean and validate data before analysis; document exclusions.
R-002	Correlations in the data are misread as clinical causation.	Medium	High	Frame insights as descriptive/diagnostic only, not clinical conclusions.
R-003	Delivery delays due to unfamiliarity with SQL or Power BI AI features.	Low	Medium	Follow a structured task sequence and use AI-assisted features to accelerate work.
R-004	Small sample sizes in some categories (e.g., certain diagnoses) skew averages.	Medium	Medium	Report sample sizes alongside averages; avoid over-generalizing.

10. Success Criteria

- BRD reviewed and approved by the project sponsor before analysis begins.
- Excel exploratory summaries (demographics, lab/medication statistics, readmission counts) completed.
- SQL schema created and all specified exploration and aggregation queries executed successfully.
- Power BI dashboard delivered with all required visuals: stacked bar chart, age-bucket column chart, multi-metric comparison by readmission status, scatter chart, and DAX-based average length-of-stay measure.
- Dashboard clearly highlights the top readmission risk factors for stakeholder review and decision-making.

11. Appendices

- Appendix A: Data Dictionary — field definitions for diabetic_data.csv.
- Appendix B: Power BI Dashboard Wireframes / Screens.
- Appendix C: BRD Template Reference — asana.com/resources/business-requirements-document-template.